

CHAPTER 6

Isomorphisms

1. Let $\phi(n) = 2n$. Then ϕ is onto since the even integer $2n$ is the image of n .
 ϕ is one-to-one since $2m = 2n$ implies that $m = n$.
 $\phi(m + n) = 2(m + n) = 2m + 2n$ so ϕ is operation preserving.
2. An automorphism of a cyclic group must carry a generator to a generator.
 Thus $1 \rightarrow 1$ and $1 \rightarrow -1$ are the only two choices for the image of 1. So let
 $\alpha : n \rightarrow n$ and $\beta : n \rightarrow -n$. Then $\text{Aut}(Z) = \{\alpha, \beta\}$.
3. ϕ is onto since any positive real number r is the image of $\sqrt{r}$. ϕ is
 one-to-one since $\sqrt{a} = \sqrt{b}$ implies that $a = b$. Finally,
 $\phi(xy) = \sqrt{xy} = \sqrt{x}\sqrt{y} = \phi(x)\phi(y)$.
4. $U(8)$ is not cyclic while $U(10)$ is.
5. Define ϕ from $U(8)$ to $U(12)$ by $\phi(1) = 1$; $\phi(3) = 5$; $\phi(5) = 7$; $\phi(7) = 11$.
 To see that ϕ is operation preserving we observe that
 $\phi(1a) = \phi(a) = \phi(a) \cdot 1 = \phi(a)\phi(1)$ for all a ;
 $\phi(3 \cdot 5) = \phi(7) = 11 = 5 \cdot 7 = \phi(3)\phi(5)$;
 $\phi(3 \cdot 7) = \phi(5) = 7 = 5 \cdot 11 = \phi(3)\phi(7)$;
 $\phi(5 \cdot 7) = \phi(3) = 5 = 7 \cdot 11 = \phi(5)\phi(7)$.
6. The identity is an isomorphism from G onto G . If β is an isomorphism
 from G onto H , then β^{-1} is an isomorphism from H onto G (see Theorem
 6.3). If β is an isomorphism from G onto H , and α is an isomorphism
 from H onto K , then $\alpha\beta$ is an isomorphism from G onto K . That $\alpha\beta$ is
 one-to-one and onto is done in Theorem 0.8. If $a, b \in G$, then
 $(\alpha\beta)(ab) = \alpha(\beta(ab)) = \alpha(\beta(a)\beta(b)) = \alpha(\beta(a))\alpha(\beta(b)) = (\alpha\beta)(a)(\alpha\beta)(b)$.
7. D_{12} has an element of order 12 and S_4 does not.
8. Properties of real numbers assure that the mapping is one-to-one and onto
 $\mathbf{R}$ and that $\log_{10}(ab) = \log_{10}(a) + \log_{10}(b)$ is a property of logs.
9. Since $T_e(x) = ex = x$ for all x , T_e is the identity. For the second part,
 observe that $T_g \circ (T_g)^{-1} = T_e = T_{gg^{-1}} = T_g \circ T_{g^{-1}}$ and cancel.
10. $\phi(na) = n\phi(a)$
11. $3\bar{a} - 2\bar{b}$.

12. Suppose α is an automorphism of G . Then $\alpha(ab) = (ab)^{-1}$ and $\alpha(ab) = \alpha(a)\alpha(b) = a^{-1}b^{-1}$. So $a^{-1}b^{-1} = (ab)^{-1} = b^{-1}a^{-1}$ for all a and b in G . Taking the inverse of both sides proves that G is Abelian.

If G is Abelian, then for all a and b in G , we have $(ab)^{-1} = (ba)^{-1} = a^{-1}b^{-1}$. Thus $\alpha(ab) = \alpha(a)\alpha(b)$.

That α is one-to-one and onto follows directly from the definitions.

13. For any x in the group, we have $(\phi_g\phi_h)(x) = \phi_g(\phi_h(x)) = \phi_g(hxh^{-1}) = ghxh^{-1}g^{-1} = (gh)x(gh)^{-1} = \phi_{gh}(x)$.
14. $\text{Aut}(Z_2) \approx \text{Aut}(Z_1) \approx Z_1$;
 $\text{Aut}(Z_6) \approx \text{Aut}(Z_4) \approx \text{Aut}(Z_3) \approx Z_2$;
 $\text{Aut}(Z_{10}) \approx \text{Aut}(Z_5) \approx Z_4$ (see Example 4 and Theorem 6.5);
 $\text{Aut}(Z_{12}) \approx \text{Aut}(Z_8)$ (see Exercise 5 and Theorem 6.5).
15. ϕ_{R_0} and $\phi_{R_{90}}$ disagree on H ; ϕ_{R_0} and ϕ_H disagree on R_{90} ; ϕ_{R_0} and ϕ_D disagree on R_{90} ; $\phi_{R_{90}}$ and ϕ_H disagree on R_{90} ; $\phi_{R_{90}}$ and ϕ_D disagree on R_{90} ; ϕ_H and ϕ_D disagree on D .
16. By Theorem 6.5, we know $|\text{Aut}(Z_6)| = |U(6)| = 2$. So $\alpha : n \rightarrow n$ and $\beta : n \rightarrow -n$ are the only two automorphisms of Z_6 .
17. We must show $\text{Aut}(G)$ has an identity, $\text{Aut}(G)$ is closed, the composition of automorphisms is associative, and the inverse of every element in $\text{Aut}(G)$ is in $\text{Aut}(G)$. Clearly, the identity function $\epsilon(x) = x$ is 1-1, onto and operation preserving. For closure let $\alpha, \beta \in \text{Aut}(G)$. That $\alpha\beta$ is 1-1 and onto follows from Theorem 0.8. For $a, b \in G$, we have $(\alpha\beta)(ab) = \alpha(\beta(ab)) = \alpha(\beta(a)\beta(b)) = (\alpha(\beta(a)))(\alpha(\beta(b))) = (\alpha\beta)(a)(\alpha\beta)(b)$. Associativity follows from properties of functions (see Theorem 0.8). Let $\alpha \in \text{Aut}(G)$. Theorem 0.8 shows that α^{-1} is 1-1 and onto. We must show that α^{-1} is operation preserving: $\alpha^{-1}(xy) = \alpha^{-1}(x)\alpha^{-1}(y)$ if and only if $\alpha(\alpha^{-1}(xy)) = \alpha(\alpha^{-1}(x)\alpha^{-1}(y))$. That is, if and only if $xy = \alpha(\alpha^{-1}(x))\alpha(\alpha^{-1}(y)) = xy$. So α^{-1} is operation preserving.
- To prove that $\text{Inn}(G)$ is a group we may use the subgroup test. Exercise 13 shows that $\text{Inn}(G)$ is closed. From $\phi_e = \phi_{gg^{-1}} = \phi_g\phi_{g^{-1}}$ we see that the inverse of ϕ_g is in $\text{Inn}(G)$. That $\text{Inn}(G)$ is a group follows from the equation $\phi_g\phi_h = \phi_{gh}$.
18. Let ϕ be an isomorphism from G to H . For any β in $\text{Aut}(G)$ define a mapping from $\text{Aut}(G)$ to $\text{Aut}(H)$ by $\Gamma(\beta) = \phi\beta\phi^{-1}$. Then Γ is 1-1 and operation preserving. (See Theorem 0.8 and Exercise 6). To see that Γ is onto observe that for any γ in $\text{Aut}(H)$, $\Gamma(\phi^{-1}\gamma\phi) = \gamma$.
19. Since $b = \phi(a) = a\phi(1)$ it follows that $\phi(1) = a^{-1}b$ and therefore $\phi(x) = a^{-1}bx$. (Here a^{-1} is the multiplicative inverse of $a \bmod n$, which exists because $a \in U(n)$.)

20. Note that ϕ must take $R_{360^\circ/n}$ to an element of order n . Since the reflections have order 2 we know $\phi(R_{360^\circ/n})$ is a rotation. Thus, $\phi(H) = \phi(\langle R_{360^\circ/n} \rangle) \subseteq K$.
21. Note that both H and K are isomorphic to the group of all permutations four symbols, which is isomorphic to S_4 . The same is true when 5 is replaced by n since both H and K are isomorphic to S_{n-1} .
22. Observe that $\langle 2 \rangle, \langle 3 \rangle, \dots$ are distinct and each is isomorphic to Z .
23. Recall when n is even, $Z(D_n) = \{R_0, R_{180}\}$. Since R_{180} and $\phi(R_{180})$ are not the identity and belong to $Z(D_n)$ they must be equal.
24. This follows directly from the subgroup tests.
25. Z_{60} contains cyclic subgroups of orders 12 and 20 and any cyclic group that has subgroups of orders 12 and 20 must be divisible by 12 and 20. So, 60 is the smallest order of any cyclic group that subgroups isomorphic to Z_{12} and Z_{60} .
26. $\phi(x) = x$; $\phi(x) = 9x$; $\phi(x) = 13x$; $\phi(x) = 17x$.
27. See Example 15 of Chapter 2.
28. It is enough to prove that the mapping is one-to-one. If $a^3 = b^3$, then $a^9 = b^9$. Now use the fact that $x^4 = 1$ for all x in $U(16)$.
29. That α is a one-to-one follows from the fact that r^{-1} exists module n . The operation preserving condition is Exercise 9 of Chapter 0.
30. The mapping $\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \rightarrow a$ is an isomorphism to Z when $a \in Z$ and to $\mathbf{R}$ when $a \in \mathbf{R}$.
31. By Part 2 of Theorem 6.2, we have $\phi(a^n) = \phi(a)^n = \gamma(a)^n = \gamma(a^n)$ thus ϕ and γ agree on all elements of $\langle a \rangle$.
32. Observe that $\phi(7) = 7\phi(1) = 13$ and since 7 is relatively prime to 50, 7^{-1} exists modulo 50. Thus, we have $\phi(1) = 7^{-1} \cdot 13 = 43 \cdot 13 = 9$ and $\phi(x) = \phi(x \cdot 1) = x\phi(1) = 9x$.
33. The inverse of a one-to-one function is one-to-one. For any $g \in G$ we have $\phi^{-1}(\phi(g)) = g$ and therefore ϕ^{-1} is onto. To verify that ϕ^{-1} is operation preserving see the answer to Exercise 15 of this chapter.
34. Since $\phi(K)$ contains $\phi(e)$, $\phi(K) \neq \emptyset$. Also, $\phi(a)(\phi(b))^{-1} = \phi(a)\phi(b^{-1}) = \phi(ab^{-1}) \in \phi(K)$.
35. $T_g(x) = T_g(y)$ if and only if $gx = gy$ or $x = y$. This shows that T_g is a one-to-one function. Let $y \in G$. Then $T_g(g^{-1}y) = y$, so that T_g is onto.

35. To prove that ϕ is 1-1 observe that $\phi(a + bi) = \phi(c + di)$ implies that $a - bi = c - di$. From properties of complex numbers this gives that $a = c$ and $b = d$. Thus $a + bi = c + di$. To prove ϕ is onto let $a + bi$ be any complex number. Then $\phi(a - bi) = a + bi$. To prove that ϕ preserves addition and multiplication note that
 $\phi((a + bi) + (c + di)) = \phi((a + c) + (b + d)i) = (a + c) - (b + d)i =$
 $(a - bi) + (c - di) = \phi(a + bi) + \phi(c + di)$. Also,
 $\phi((a + bi)(c + di)) = \phi((ac - bd) + (ad + bc)i) = (ac - bd) - (ad + bc)i$ and
 $\phi(a + bi)\phi(c + di) = (a - bi)(c - di) = (ac - bd) - (ad + bc)i$.
36. $U(20)$ has three elements of order 2 whereas $U(24)$ has seven.
37. To prove that ϕ is 1-1 observe that $\phi(a + bi) = \phi(c + di)$ implies that $a - bi = c - di$. From properties of complex numbers this gives that $a = c$ and $b = d$. Thus $a + bi = c + di$. To prove ϕ is onto let $a + bi$ be any complex number. Then $\phi(a - bi) = a + bi$. To prove that ϕ preserves addition and multiplication note that
 $\phi((a + bi) + (c + di)) = \phi((a + c) + (b + d)i) = (a + c) - (b + d)i =$
 $(a - bi) + (c - di) = \phi(a + bi) + \phi(c + di)$. Also,
 $\phi((a + bi)(c + di)) = \phi((ac - bd) + (ad + bc)i) = (ac - bd) - (ad + bc)i$ and
 $\phi(a + bi)\phi(c + di) = (a - bi)(c - di) = (ac - bd) - (ad + bc)i$.
38. Map $a + b\sqrt{2} \rightarrow \begin{bmatrix} a & 2b \\ b & a \end{bmatrix}$. This map preserves both addition and multiplication.
39. First observe the Z is a cyclic group generated by 1. By property 3 of Theorem 6.3, it suffices to show that Q is not cyclic under addition. By way of contradiction suppose that $Q = \langle p/q \rangle$. But then $p/2q$ is a rational number that is not in $\langle p/q \rangle$.
40. S_8 contains $\langle (12345)(678) \rangle$ which has order 15. Since $|U(16)| = 8$, By Cayley's Theorem S_8 contains a subgroup isomorphic to $U(16)$. The elements of D_8 can be represented as permutations of the 8 vertices of a regular 8-gon.
41. The notation itself suggests that

$$\phi(a + bi) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

is the appropriate isomorphism. To verify this note that

$$\phi((a + bi) + (c + di)) = \begin{bmatrix} a + c & -(b + d) \\ (b + d) & a + c \end{bmatrix} =$$

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} + \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \phi(a + bi) + \phi(c + di).$$

$$\begin{aligned} \text{Also, } \phi((a+bi)(c+di)) &= \phi((ac-bd) + (ad+bc)i) = \\ &= \begin{bmatrix} (ac-bd) & -(ad+bc) \\ (ad+bc) & ac-bd \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \\ &= \phi(a+bi)\phi(c+di). \end{aligned}$$

42. $\phi((a_1, \dots, a_n) + (b_1, \dots, b_n)) = (-a_1, \dots, -a_n) = (-b_1, \dots, -b_n)$ implies $(a_1, \dots, a_n) = (b_1, \dots, b_n)$ so that ϕ is 1-1. For any $(a_1, \dots, a_n)$, we have $\phi(-a_1, \dots, -a_n) = (a_1, \dots, a_n)$ so ϕ is onto.
 $\phi((a_1 + b_1, \dots, a_n + b_n)) = (-(a_1 + b_1), \dots, -(a_n + b_n)) = (-a_1, \dots, -a_n)(-b_1, \dots, -b_n) = \phi((a_1, \dots, a_n)) + \phi((b_1, \dots, b_n))$. ϕ reflects each point through the origin.
43. Yes, by Cayley's Theorem.
44. $(ab)^2 = a^2b^2$ shows that the mapping is O.P. To show it is 1-1, note that $a^2 = b^2$, implies $e = a^2b^{-2} = (ab^{-1})^2$ so that $|ab^{-1}| = 1$ or 2 . Thus, $a = b$. Since G is finite, 1-1 implies onto. For Z under addition, $g \rightarrow 2g$ is not onto but is 1-1 and operation preserving.
45. Observe that $\phi_g(y) = gyg^{-1}$ and $\phi_{zg}(y) = zgy(zg)^{-1} = zgyg^{-1}z^{-1} = gyg^{-1}$, since $z \in Z(G)$. So, $\phi_g = \phi_{zg}$.
46. In $\mathbf{R}$ under addition every nonzero element has infinite order. In $\mathbf{R}^*$ under multiplication -1 has order 2.
47. $\phi_g = \phi_h$ implies $gxg^{-1} = h x h^{-1}$ for all x . This implies $h^{-1}g x (h^{-1}g)^{-1} = x$, and therefore $h^{-1}g \in Z(G)$.
48. $\phi_g = \phi_h$ if and only if $h^{-1}g \in Z(G)$.
49. By Exercise 47 $\phi_\alpha = \phi_\beta$ implies $\beta^{-1}\alpha$ is in $Z(S_n)$ and by Exercise 58 in Chapter 5, $Z(S_n) = \{\epsilon\}$.
50. Since 2 is not in the image the mapping is not onto.
51. Since both ϕ and γ both take e to itself, H is not empty. Assume a and b belongs to H . Then $\phi(ab^{-1}) = \phi(a)\phi(b^{-1}) = \phi(a)\phi(b)^{-1} = \gamma(a)\gamma(b)^{-1} = \gamma(a)\gamma(b^{-1}) = \gamma(ab^{-1})$. Thus ab^{-1} is in H .
52. G is Abelian.
53. Since $\phi(e) = e = e^{-1}$, H is not empty. Assume that a and b belong to H . Then $\phi(ab) = \phi(a)\phi(b) = a^{-1}b^{-1} = b^{-1}a^{-1} = (ab)^{-1}$, and H is closed under multiplication. Moreover, because $\phi(a^{-1}) = \phi(a)^{-1} = (a^{-1})^{-1}$ we have that H is closed under inverses.
54. Since $|R_{45}| = 8$ it must map to elements of order 8. Since the integers between 1 and 8 relatively prime to 8 are 1, 3, 5, 7 the elements of order 8 are $R_{45}, R_{45}^3, R_{45}^5, R_{45}^7$.

55. Since -1 is the unique element of $\mathbf{C}^*$ of order 2, $\phi(-1) = -1$. Since i and $-i$ are the only elements of $\mathbf{C}^*$ of order 4, $\phi(i) = i$ or $-i$.
56. The mapping $\phi(x) = (3/2)x$ is an isomorphism from G onto H .
Multiplication is not preserved. When $G = \langle m \rangle$ and $H = \langle n \rangle$ the mapping $\phi(x) = (n/m)x$ is an isomorphism from G onto H .
57. Z_{120}, D_{60}, S_5 . Z_{120} is Abelian, the other two are not. D_{60} has an element of order 60 and S_5 does not.
58. Using Exercise 23 we have
 $\phi(V) = \phi(R_{180}H) = \phi(R_{180})\phi(H) = R_{180}D = D'$.
59. Observe that $\phi(D) = \phi(R_{90}V) = \phi(R_{90})\phi(V) = R_{270}V = D'$ and
 $\phi(H) = \phi(R_{90}D) = \phi(R_{90})\phi(D) = R_{270}D' = H$.
60. $\alpha_5 = (0)(157842)(36); \alpha_8 = (0)(18)(27)(36)(45)$.
61. $(R_0R_{90}R_{180}R_{270})(HD'VD)$.
62. By part 2 of Theorem 6.2, $n\phi(1/n) = \phi(1)$ so that $\phi(1/n) = (1/n)\phi(1)$.
Also, by Part 2 of Theorem 6.2,
 $\phi(m/n) = m\phi(1/n) = m(1/n)\phi(1) = (m/n)\phi(1)$.
63. The mapping $\phi(x) = x^2$ is one-to-one from Q^+ to Q^+ since $a^2 = b^2$ implies $a = b$ when both a and b are positive. Moreover, $\phi(ab) = \phi(a)\phi(b)$ for all a and b . However, ϕ is not onto since there is no rational whose square is 2. So, the image of ϕ is a proper subgroup of Q^+ .
64. The argument given in Exercise 40 shows that an isomorphic image of Q has the form aQ where a is a nonzero rational. But $aQ = Q$.
65. Suppose that ϕ is an automorphism of R^* and a is positive. Then
 $\phi(a) = \phi(\sqrt{a}\sqrt{a}) = \phi(\sqrt{a})\phi(\sqrt{a}) = \phi(\sqrt{a})^2 > 0$. Now suppose that a is negative but $\phi(a) = b$ is positive. Then by the case we just did
 $a = \phi^{-1}(\phi(a)) = \phi^{-1}(b)$ is positive. This is a contradiction.
66. If ϕ were an isomorphism then
 $0 = \phi(1) = \phi(-1 \cdot -1) = \phi(-1) + \phi(-1) = 2\phi(-1)$ implies that $\phi(-1) = 0$.
But then ϕ is not 1-1.
67. Say ϕ is an isomorphism from Q to $\mathbf{R}^+$ and ϕ takes 1 to a . It follows that the integer r maps to a^r . Then $a = \phi(1) = \phi(s \frac{1}{s}) = \phi(\frac{1}{s} + \cdots + \frac{1}{s}) = \phi(\frac{1}{s})^s$ and therefore $a^{\frac{1}{s}} = \phi(\frac{1}{s})$. Thus, the rational r/s maps to $a^{r/s}$. But $a^{r/s} \neq a^\pi$ for any rational number r/s .

CHAPTER 7

Cosets and Lagrange's Theorem

1. $H, 1 + H, 2 + H$. To see that there are no others notice that for any integer n we can write $n = 3q + r$ where $0 \leq r < 3$. So, $n + H = r + 3q + H = r + H$, where $r = 0, 1$ or 2 .
2. $b - a \in H$
3. **a.** $11 + H = 17 + H$ because $17 - 11 = 6$ is in H ;
 b. $-1 + H = 5 + H$ because $5 - (-1) = 6$ is in H ;
 c. $7 + H \neq 23 + H$ because $23 - 7 = 16$ is not in H .
4. $0 + \langle n \rangle, 1 + \langle n \rangle, \dots, n - 1 + \langle n \rangle; n$
5. Since $8/2 = 4$, there are four cosets. Let $H = \{1, 11\}$. The cosets are $H, 7H, 13H, 19H$.
6. $|\langle a^5 \rangle| = 3$ so there are $15/3 = 5$ cosets. They are $\langle a^5 \rangle, a\langle a^5 \rangle, a^2\langle a^5 \rangle, a^3\langle a^5 \rangle, a^4\langle a^5 \rangle$.
7. Since $|a^4| = 15$, there are two cosets: $\langle a^4 \rangle$ and $a\langle a^4 \rangle$.
8. Let F and F' be distinct reflections in D_3 . Then take $H = \{R_0, F\}$ and $K = \{R_0, F'\}$.
9. $H = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$, $\alpha_5 H = \{\alpha_5, \alpha_8, \alpha_6, \alpha_7\}$, $\alpha_9 H = \{\alpha_9, \alpha_{11}, \alpha_{12}, \alpha_{10}\}$. There are $24/4 = 6$ left cosets.
10. Observe that $aH = bK$ implies that $b^{-1}aH = K$. Thus $b^{-1}aH$ is a subgroup of G . From part 9 of the lemma in Chapter 7 we have that $b^{-1}a$ is in H and therefore $b^{-1}aH = H$.
11. Let ga belong to $g(H \cap K)$ where a is in $H \cap K$. Then by definition ga is in $gH \cap gK$. Now let $x \in gH \cap gK$. Then $x = gh$ for some $h \in H$ and $x = gk$ for some $k \in K$. Cancellation then gives $h = k$. Thus $x \in g(H \cap K)$.
12. By Lagrange's Theorem $|H| = 5, 31$ or 155 . But $|H| = 5$ implies that all non-identity elements in H have order 5 and $|H| = 31$ implies that all non-identity elements in H have order 31.
13. Suppose that $h \in H$ and $h < 0$. Then $h\mathbf{R}^+ \subseteq hH = H$. But $h\mathbf{R}^+$ is the set of all negative real numbers. Thus $H = \mathbf{R}^*$.
14. The coset containing $c + di$ is the circle with center at the origin and radius $\sqrt{c^2 + d^2}$.

15. By Lagrange's Theorem the possible orders are 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60.
16. 84 or 210.
17. By Lagrange's Theorem, the only possible orders for the subgroups are 1, p and q . By Corollary 3 of Lagrange's Theorem, groups of prime order are cyclic. The subgroup of order 1 is $\langle e \rangle$.
18. Note that $\phi(n) = |U(n)|$ then use Corollary 4 of Lagrange's Theorem and mimic the proof of Corollary 5 of Lagrange's Theorem.
19. By Exercise 18 we have $5^6 \bmod 7 = 1$. So, using mod 7 we have $5^{15} = 5^6 \cdot 5^6 \cdot 5^2 \cdot 5 = 1 \cdot 1 \cdot 4 \cdot 5 = 6$; $7^{13} \bmod 11 = 2$.
20. Note that $n - 1 \in U(n)$ and has order 2.
21. By Corollary 4 of Theorem 7.1 $g^n = e$. Then since $g^m = e$ and $g^n = e$ we know that $|g|$ is a common divisor of both m and n . So, $|g| = 1$.
22. Since $|H \cap K|$ must divide 12 and 35, $|H \cap K| = 1$. If H and K are relatively prime, $|H \cap K| = 1$.
23. First observe that for all $n \geq 3$ the subgroup of rotations of D_n is isomorphic to Z_n . If n is even let F be any reflection in D_n . Then the set $\{R_0, R_{180}, F, FR_{180}\}$ is closed and therefore a subgroup of order 4. Now suppose that D_n has a subgroup K of order 4. If K is cyclic then it has a rotation of order 4 and therefore 4 divides n . If K is not cyclic, then it has three elements of order 2. Since there is only one rotation of order 2, K must contain two reflections F_1 and F_2 . But then F_1F_2 is a rotation and has order 2 so n is even.
24. Since $U(p)$ is a cyclic group of order $p - 1$ there is an element a in $U(p)$ with $|a| = p - 1$. By our assumption we have $a^k = a$, which implies that $a^{k-1} = 1$. So, by Corollary 2 of Theorem 4.1, we have $p - 1$ divides $k - 1$.
25. Since G has odd order, no element can have order 2. Thus, for each $x \neq e$, we know that $x \neq x^{-1}$. So, we can write the product of all elements in the form $ea_1a_1^{-1}a_2a_2^{-1} \cdots a_na_n^{-1} = e$.
26. First suppose G is infinite. Let $x \in G, x \neq e$. Then $G = \langle x \rangle$ and $\langle x^2 \rangle$ contradicts the hypothesis. Next assume G is finite and $e \neq x \in G$. Then $\langle x \rangle = G$ (otherwise $\langle x \rangle$ is nontrivial and proper). By Theorem 4.3 G has a subgroup for each divisor of $|G|$ and since the only subgroups of G have orders $|G|$ and 1 we have that only divisors of $|G|$ are $|G|$ and 1. So, $|G|$ is prime.
27. Let H be the subgroup of order p and K be the subgroup of order q . Then $H \cup K$ has $p + q - 1 < pq$ elements. Let a be any element in G that is not in $H \cup K$. By Lagrange's Theorem, $|a| = p, q$, or pq . But $|a| \neq p$, for if so, then $\langle a \rangle = H$. Similarly, $|a| \neq q$.

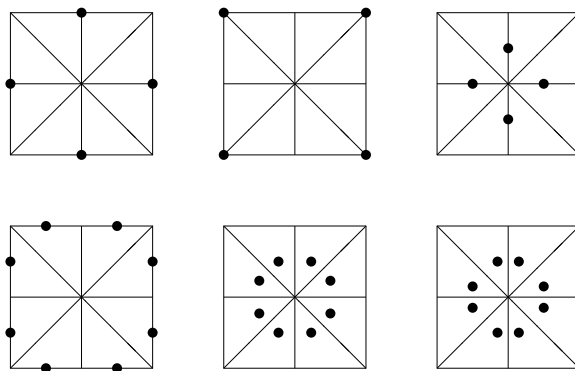
28. Let $e \neq g \in G$. Then $|g| = 5$ or 25 . If $|g| = 25$ for some g , then G is cyclic. If there is no such g , then $|g| = 1$ or 5 for all g .
29. The possible orders are $1, 3, 11, 33$. If $|x| = 33$, then $|x^{11}| = 3$ so we may assume that there is no element of order 33 . By the Corollary of Theorem 4.4, the number of elements of order 11 is a multiple of 10 so they account for $0, 10, 20$, or 30 elements of the group. The identity accounts for one more. So, at most we have accounted for 31 elements. By Corollary 2 of Lagrange's Theorem, the elements unaccounted for have order 3 .
30. Let $e \neq g \in G$. If $|g| = 8$, $|g^4| = 2$. If $|g| = 4$, $|g^2| = 2$.
31. No. By Lagrange's Theorem, the elements of a group of order 55 must have orders $1, 5, 11$, or 55 . By Theorem 4.4 a cyclic group of order 55 cannot have exactly 20 elements of order 11 . So, a group with exactly 20 elements of order 11 must have exactly 34 elements of order 5 . This contradicts the Corollary to Theorem 4.4.
32. For any positive integer n let $\omega_n = \cos\left(\frac{2\pi}{n}\right) + i \sin\left(\frac{2\pi}{n}\right)$. The finite subgroups of C^* are those of the form $\langle \omega_n \rangle$. To verify this, let H denote any finite subgroup of C^* of order n . Then every element of H is a solution to $x^n = 1$. But the solution set of $x^n = 1$ in C^* is $\langle \omega_n \rangle$.
33. Observe that $|G : H| = |G|/|H|$, $|G : K| = |G|/|K|$, $|K : H| = |K|/|H|$. So, $|G : K||K : H| = |G|/|H| = |G : H|$.
34. 2520
35. Since the reflections in a dihedral group have order 2 , the generators of the subgroups of orders 12 and 20 must be rotations. The smallest rotation subgroup of a dihedral group that contains rotations of orders 12 and 20 must have order divisible by 12 and 20 and therefore must be a multiple of 60 . So, D_{60} is the smallest such dihedral group.
36. Since $|g|$ must divide both 14 and 21 , $|g| = 1$ or 7 .
37. Let a have order 3 and b be an element of order 3 not in $\langle a \rangle$. Then $\langle a \rangle \langle b \rangle = \{a^i b^j \mid i = 0, 1, 2, j = 0, 1, 2\}$ is a subgroup of G of order 9 . Now use Lagrange's Theorem.
38. Suppose that $|G|/|Z(G)| = p$ where p is prime. Let $a \in G$, but $a \notin Z(G)$. Since the subgroup $C(a)$ contains both a and $Z(G)$ we know that $Z(G)$ is proper subgroup of $C(a)$. Since $p = (|G|/|C(a)|)(|C(a)|/|Z(G)|)$ we have $|G|/|C(a)| = 1$ and therefore $C(a) = G$ and $a \in Z(G)$.
39. Since $|H \cap K|$ is a common divisor of 24 and 20 it must divide 4 . But groups of orders $1, 2$ and 4 are Abelian.

40. Suppose G is a group of order 63. Let a be any non-identity element in G . By Lagrange's Theorem, $|a| = 63, 21, 9, 7$, or 3 . If $|a| = 3$, we are done. If $|a| = 63$, then $|a^{21}| = 3$; if $|a| = 21$, then $|a^7| = 3$; and if $|a| = 9$, then $|a^3| = 3$. So, if any of these cases occur we are done. Thus we may assume that all 62 non-identity elements in G have order 7. But by the Corollary to Theorem 4.4, the number of elements of order 7 must be a multiple of 6.
41. Let $a \in G$ and $|a| = 5$. Then by Theorem 7.2 we know that the set $\langle a \rangle H$ has exactly $5 \cdot |H|/|\langle a \rangle \cap H|$ elements and $|\langle a \rangle \cap H|$ divides $|\langle a \rangle| = 5$. It follows that $|\langle a \rangle \cap H| = 5$ and therefore $\langle a \rangle \cap H = \langle a \rangle$.
42. Suppose the $a^k = b^k$. By the Corollary to Theorem 0.2 there are integers s and t such that $1 = ns + kt$. Then by Corollary 4 of Lagrange's Theorem we have $a = a^{ns+kt} = (a^n)^s (a^k)^t = (a^k)^t = (b^n)^s (b^k)^t = b^{ns+kt} = b$. To prove that the mapping is an automorphism when the group is also Abelian note that by Exercise 10 of Chapter 5 a 1-1 mapping from a finite set to itself is onto. Lastly, observe that $(ab)^k = a^k b^k$.
43. Certainly, $a \in \text{orb}_G(a)$. Now suppose $c \in \text{orb}_G(a) \cap \text{orb}_G(b)$. Then $c = \alpha(a)$ and $c = \beta(b)$ for some α and β , and therefore $(\beta^{-1}\alpha)(a) = \beta^{-1}(\alpha(a)) = \beta^{-1}(c) = b$. So, if $x \in \text{orb}_G(b)$, then $x = \gamma(b) = \gamma(\beta^{-1}\alpha)(a) = (\gamma\beta^{-1}\alpha)(a)$. This proves $\text{orb}_G(b) \subseteq \text{orb}_G(a)$. By symmetry, $\text{orb}_G(a) \subseteq \text{orb}_G(b)$.
44. Since reflections have order 2 the subgroup must consist entirely of rotations and the subgroup of all rotations is cyclic.
45. **a.** $\text{stab}_G(1) = \{(1), (24)(56)\}$; $\text{orb}_G(1) = \{1, 2, 3, 4\}$
b. $\text{stab}_G(3) = \{(1), (24)(56)\}$; $\text{orb}_G(3) = \{3, 4, 1, 2\}$
c. $\text{stab}_G(5) = \{(1), (12)(34), (13)(24), (14)(23)\}$; $\text{orb}_G(5) = \{5, 6\}$
46. Let $|G| = 12$ and let $a \in G$ be a nonidentity element. By Lagrange's Theorem, $|a| = 12, 6, 4, 3$, or 2 . If $|a| = 12$, then $|a^6| = 2$. Similarly, if a has order 6 or 4 then there is an element of order 2. So, we may assume that all 11 nonidentity elements have order 3. But elements of order 3 come in pairs (if $|a| = 3$, then $|a^2| = 3$). Since this is a contradiction, one of the earlier cases must occur.
47. Consider the mapping from G to G defined by $\phi(x) = x^2$. To prove that it is one-to-one assume that $x^2 = y^2$ and let $|G| = 2k + 1$. Then $x = xe = xx^{2k+1} = x^{2k+2} = (x^2)^{k+1} = (y^2)^{k+1} = y^{2k+2} = yy^{2k+1} = ye = y$. By Exercise 12 of Chapter 5, ϕ is also onto.
48. It follows from Lagrange's Theorem that $|H \cap K| = 1$ or q and Theorem 7.2 rules out $|H \cap K| = 1$.
49. Use Theorem 7.2.

50. Observe that $|\text{orb}_{A_5}(5)| = 5$. Now use the Orbit-Stabilizer Theorem to show that $|\text{stab}_{A_5}(5)| = 12$. Note that the same argument applies to $\text{stab}_{A_5}(i)$ for $i = 1, 2, 3$, and 4 .
51. Suppose that H is a subgroup of A_5 of order 30. We claim that H contains all 20 elements of A_5 that have order 3. To verify this assume that there is some α in A_5 of order 3 that is not in H . Then $A_5 = H \cup \alpha H$. It follows that $\alpha^2 H = H$ or $\alpha^2 = \alpha H$. Since the latter implies that $\alpha \in H$, we have that $\alpha^2 H = H$, which implies that $\alpha^2 \in H$. But then $\langle \alpha \rangle = \langle \alpha^2 \rangle \subseteq H$, which is a contradiction of our assumption that α is not in H . The same argument shows that H must contain all 24 elements of order 5. Since $|H| = 30$ we have a contradiction.
52. Suppose that H is a subgroup of A_5 of order 20. We claim that H contains all 24 elements of A_5 that have order 5. To verify this assume that there is some α in A_5 of order 5 that is not in H . Then $A_5 = H \cup \alpha H \cup \alpha^2 H$. To see that the coset $\alpha^2 H$ is not the same as H note that $\alpha^2 H = H$ implies that $\langle \alpha^2 \rangle \subset H$ and $\langle \alpha \rangle = \langle \alpha^2 \rangle$. Moreover, $\alpha^2 H$ is not the same as αH for then $\alpha \in H$. It follows that $\alpha^3 H$ is equal to one of the cosets H , αH or $\alpha^2 H$. If $\alpha^3 H = H$ then $\alpha^3 \in H$ and therefore $\langle \alpha \rangle = \langle \alpha^3 \rangle \subseteq H$, which contradicts the assumption that α is not in H . If $\alpha^3 H = \alpha H$ then $\alpha^2 \in H$ and therefore $\langle \alpha \rangle = \langle \alpha^2 \rangle \subseteq H$, which contradicts the assumption that α is not in H . If $\alpha^3 H = \alpha^2 H$ then $\alpha \in H$ which contradicts the assumption that α is not in H . The same argument shows that H must contain all 24 elements of order 5. Since $|H| = 20$ we have a contradiction. An analogous argument shows that A_5 has no subgroup of order 15.
53. Observe that $\alpha(a_i) = a_{i+1}$, $\alpha^2(a_i) = a_{i+2}, \dots$, $\alpha^k(a_i) = a_i$ where all subscripts are taken mod k .
54. If G has an element of order 105 then G is cyclic. So we may assume that no element has order 105. Then by Lagrange every element in G has order 1, 3, 5, 7, 15, 21 or 35. If $|a| = 35$, then $\langle a \rangle$ contains all elements of G of order 35. So the maximum number of elements in G of order 35 is $\phi(35) = \phi(5)\phi(7) = 24$. Similarly, the maximum number of elements of orders 21, 15, 7, 5, 3, and 1 is 14, 8, 6, 4, 2, and 1. But $24 + 14 + 8 + 6 + 4 + 2 + 1 < 105$.
55. Suppose that H is a subgroup of S_5 of order 60. An argument analogous to that given in Exercise 51 in this chapter shows that H must contain all 24 elements in S_5 of order 5 and all 20 elements in S_5 of order 3. Since these 44 elements are also in A_5 we know that $|A_5 \cap H|$ divides 60 and is greater than 30. So, $H = A_5$.
56. If $|Z(A_4)| > 1$, then A_4 would have an element of order 2 or order 3 that commutes with every element. But any subgroup generated by an element of order 2 and an element of order 3 that commute has order 6. This

contradicts the fact shown in Example 5 that A_4 has no subgroup of order 6.

57. Suppose that $B \in G$ and $\det B = 2$. Then $\det (A^{-1}B) = 1$, so that $A^{-1}B \in H$ and therefore $B \in AH$. Conversely, for any $Ah \in AH$ we have $\det Ah = (\det A)(\det h) = 2 \cdot 1 = 2$.
58. The circle passing through Q , with center at P .
59. It is the set of all permutations that carry face 2 to face 1.
- 60.



$$\{R_0, H\}; \{R_0, D'\}; \{R_0, H\} \\ \{R_0\}; \{R_0\}; \{R_0\}.$$

61. If $aH = bH$, then $b^{-1}a \in H$. So $\det (b^{-1}a) = (\det b^{-1})(\det a) = (\det b^{-1})(\det a) = (\det b)^{-1}(\det a) = 1$. Thus $\det a = \det b$. Conversely, we can read this argument backwards to get that $\det a = \det b$ implies $aH = bH$.
62. a 12 b 24 c 60 d 60
63. To prove that the set is closed note that $\alpha\beta^2 = (13) = \beta^2\alpha^3$, $\alpha^2\beta^2 = (14)(23) = \beta^2\alpha^2$, and $\alpha^3\beta^2 = (24) = \beta^2\alpha$.
64. The order of the symmetry group would have to be $6 \cdot 20 = 120$.
65. Since the order of G is divisible by both 10 and 25 it must be divisible by 50. But the only number less than 100 that is divisible by 50 is 50.

CHAPTER 8

External Direct Products

1. Closure and associativity in the product follows from the closure and associativity in each component. The identity in the product is the n -tuple with the identity in each component. The inverse of $(g_1, g_2, \dots, g_n)$ is $(g_1^{-1}, g_2^{-1}, \dots, g_n^{-1})$.
2. In general, $(1, 1, \dots, 1)$ is an element of largest order in $Z_{n_1} \oplus Z_{n_2} \oplus \dots \oplus Z_{n_t}$. To see this note that because the order of the 1 in each component is the order of the group in that component, $|(1, 1, \dots, 1)| = \text{lcm}(n_1, n_2, \dots, n_t)$ and the order of every element in the product must divide $\text{lcm}(n_1, n_2, \dots, n_t)$.
3. The mapping $\phi(g) = (g, e_H)$ is an isomorphism from G to $G \oplus \{e_H\}$. To verify that ϕ is one-to-one, we note that $\phi(g) = \phi(g')$ implies $(g, e_H) = (g', e_H)$ which means that $g = g'$. The element $(g, e_H) \in G \oplus \{e_H\}$ is the image of g . Finally, $\phi((g, e_H)(g', e_H)) = \phi((gg', e_H)) = gg' = \phi((g, e_H))\phi((g', e_H))$. A similar argument shows that $\phi(h) = (e_G, h)$ is an isomorphism from H onto $\{e_G\} \oplus H$.
4. $(g, h)(g', h') = (g', h')(g, h)$ for all g, g', h, h' if and only if $gg' = g'g$ and $hh' = h'h$, that is, if and only if G and H are Abelian. A corresponding statement holds for the external direct product of any number of groups.
5. If $Z \oplus Z = \langle(a, b)\rangle$ then neither a nor b is 0. But then $(1, 0) \notin \langle(a, b)\rangle$. $Z \oplus G$ is not cyclic when $|G| > 1$.
6. $Z_8 \oplus Z_2$ contains elements of order 8, while $Z_4 \oplus Z_4$ does not.
7. Define a mapping from $G_1 \oplus G_2$ to $G_2 \oplus G_1$ by $\phi(g_1, g_2) = (g_2, g_1)$. To verify that ϕ is one-to-one, we note that $\phi((g_1, g_2)) = \phi((g'_1, g'_2))$ implies $(g_2, g_1) = (g'_2, g'_1)$. From this we obtain that $g_1 = g'_1$ and $g_2 = g'_2$. The element (g_2, g_1) is the image of (g_1, g_2) so ϕ is onto. Finally, $\phi((g_1, g_2)(g'_1, g'_2)) = \phi((g_1g'_1, g_2g'_2)) = (g_2g'_2, g_1g'_1) = (g_2, g_1)(g'_2, g'_1) = \phi((g_1, g_2))\phi((g'_1, g'_2))$. In general, the external direct product of any number of groups is isomorphic to the external direct product of any rearrangement of those groups.
8. No, $Z_3 \oplus Z_9$ does not have an element of order 27. See also Theorem 8.2.
9. In $Z_6 \oplus Z_2$, $|\langle(1, 0)\rangle| = 6$ and $|\langle(1, 1)\rangle| = 6$.

10. Z_9 has 6 elements of order 9 (the members of $U(9)$). Any of these together with any element of Z_3 gives an ordered pair whose order is 9. So $Z_3 \oplus Z_9$ has 18 elements of order 9.
11. In both $Z_4 \oplus Z_4$ and $Z_{8000000} \oplus Z_{4000000}$, $|(a, b)| = 4$ if and only if $|a| = 4$ and $|b| = 1, 2$ or 4 or if $|b| = 4$ and $|a| = 1$ or 2 (we have already counted the case that $|a| = 4$). For the first case, we have $\phi(4) = 2$ choices for a and $\phi(4) = \phi(2) + \phi(1) = 4$ choices for b to give us 8 in all. For the second case, we have $\phi(4) = 2$ choices for b and $\phi(2) + \phi(1) = 2$ choices for a . This gives us a total of 12.
In the general case observe that by Theorem 4.4 that as long as d divides n the number of elements of order d in a cyclic group depends only on d .
12. $Z_{12}, Z_6 \oplus Z_2, D_6, A_4$. The first two are Abelian and the second two are not. Z_{12} is cyclic and $Z_6 \oplus Z_2$ is not. D_6 has an element of order 6 and A_4 does not.
13. Z_{n^2} and $Z_n \oplus Z_n$.
14. The group of rotations is Abelian and a group of order 2 is Abelian; now use Exercise 4.
15. Define a mapping ϕ from $\mathbf{C}$ to $\mathbf{R} \oplus \mathbf{R}$ by $\phi(a + bi) = (a, b)$. To verify that ϕ is one-to-one note that $\phi(a + bi) = \phi(a' + b'i)$ implies that $(a, b) = (a', b')$. So, $a = a'$ and $b = b'$ and therefore $a + bi = a' + b'i$. The element (a, b) in $\mathbf{R} \oplus \mathbf{R}$ is the image of $a + bi$ so ϕ is onto. Finally, $\phi((a + bi) + (a' + b'i)) = \phi((a + a') + (b + b')i) = (a + a', b + b') = (a, b) + (a', b') = \phi(a + bi) + \phi(a' + b'i)$.
16. Let $\alpha : G_1 \rightarrow G_2$ and $\beta : H_1 \rightarrow H_2$ be isomorphisms. Then $\gamma : G_1 \oplus H_1 \rightarrow G_2 \oplus H_2$ given by $\gamma((g_1, h_1)) = (\alpha(g_1), \beta(h_1))$ is an isomorphism. A corresponding statement holds for the external direct product of any number of groups.
17. By Exercise 3 in this chapter G is isomorphic to $G \oplus \{e_H\}$ and H is isomorphic to $\{e_G\} \oplus H$. Since subgroups of cyclic groups are cyclic, we know that $G \oplus \{e_H\}$ and $\{e_G\} \oplus H$ are cyclic. In general, if the external direct product of any number of groups is cyclic, each of the factors is cyclic.
18. $\langle (10, 10) \rangle; \langle 20 \rangle \oplus \langle 5 \rangle$.
19. $\langle m/r \rangle \oplus \langle n/s \rangle$.
20. Observe that $Z_9 \oplus Z_4 \approx Z_4 \oplus Z_9 \approx \langle 3 \rangle \oplus \langle 2 \rangle$.
21. Since $\langle (g, h) \rangle \subseteq \langle g \rangle \oplus \langle h \rangle$, a necessary and sufficient condition for equality is that $\text{lcm}(|g|, |h|) = |(g, h)| = |\langle g \rangle \oplus \langle h \rangle| = |g||h|$. This is equivalent to $\text{gcd}(|g|, |h|) = 1$.

22. 48; 6

23. In the general case there are $(3^n - 1)/2$.

24. Observe that $|(a, b)| = 2$ if and only if $|a| = 1$ or 2 and $|b| = 1$ or 2 but not both $|a| = 1$ and $|b| = 1$. So, there are $(m + 2)(n + 1) - 1 = mn + m + 2n + 1$ elements of order 2.

25. Define a mapping ϕ from M to N by $\phi \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = (a, b, c, d)$. To verify that ϕ is one-to-one we note that $\phi \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = \phi \left(\begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} \right)$ implies $(a, b, c, d) = (a', b', c', d')$. Thus $a = a', b = b', c = c'$, and $d = d'$. This proves that ϕ is one-to-one. The element (a, b, c, d) is the image of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ so ϕ is onto. Finally,

$$\phi \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} \right) = \phi \left(\begin{bmatrix} a + a' & b + b' \\ c + c' & d + d' \end{bmatrix} \right) = (a + a', b + b', c + c', d + d') = (a, b, c, d) + (a', b', c', d') = \phi \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) + \phi \left(\begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} \right).$$

Let $\mathbf{R}^k$ denote $\mathbf{R} \oplus \mathbf{R} \oplus \cdots \oplus \mathbf{R}$ (k factors). Then the group of $m \times n$ matrices under addition is isomorphic to R^{mn} .

26. D_6 . Since $S_3 \oplus Z_2$ is non-Abelian, it must be isomorphic to A_4 or D_6 . But $S_3 \oplus Z_2$ contains an element of order 6 and A_4 does not.

27. Since $(g, g)(h, h)^{-1} = (gh^{-1}, gh^{-1})$, H is a subgroup. When $G = \mathbf{R}$, $G \oplus G$ is the plane and H is the line $y = x$.

28. $D_{12}, S_4, A_4 \oplus Z_2, D_4 \oplus Z_3, D_3 \oplus Z_4, D_3 \oplus Z_2 \oplus Z_2$.

29. $\langle (3, 0) \rangle, \langle (3, 1) \rangle, \langle (3, 2) \rangle, \langle (0, 1) \rangle$

30. $\langle (1, 0) \rangle, \langle (1, 1) \rangle, \langle (1, 2) \rangle, \langle (1, 3) \rangle, \langle (0, 1) \rangle, \langle (2, 1) \rangle, \{(0, 0), (2, 0), (0, 2), (2, 2)\}$

31. $\text{lcm}(6, 10, 15) = 30$; $\text{lcm}(n_1, n_2, \dots, n_k)$.

32. In general, if m and n are even, then $Z_m \oplus Z_n$ has exactly 3 elements of order 2. For if $|(a, b)| = 2$, then $|a| = 1$ or 2 and $|b| = 1$ or 2 but not both a and b have order 1. Since any cycle group of even order has exactly 1 element of order 2 and 1 of order 1 there are only 3 choices for (a, b) .

33. $\langle 400 \rangle \oplus \langle 50 \rangle = \{0, 400\} \oplus \{0, 50, 100, 150\}$

34. $\langle 4 \rangle \oplus \langle 0 \rangle \oplus \langle 5 \rangle$

35. In $\mathbf{R}^* \oplus \mathbf{R}^*$ $(1, -1)$, $(-1, 1)$ and $(-1, -1)$ have order 2 whereas in $\mathbf{C}^*$ the only element of order 2 is -1 . But isomorphisms preserve order.

36. $Z_3 \oplus Z_3$

37. Define the mapping from G to $Z \oplus Z$ by $\phi(3^m 6^n) = (m, n)$. To verify that ϕ is one-to-one note that $\phi(3^m 6^n) = \phi(3^s 6^t)$ implies that $(m, n) = (s, t)$, which in turn implies that $m = s$ and $n = t$. So, $3^m 6^n = 3^s 6^t$. The element (m, n) is the image of $3^m 6^n$ so ϕ is onto. Finally, $\phi((3^m 6^n)(3^s 6^t)) = \phi(3^{m+s} 6^{n+t}) = (m+s, n+t) = (m, n) + (s, t) = \phi(3^m 6^n) + \phi(3^s 6^t)$ shows that ϕ is operation preserving. When $G = \{3^m 9^n \mid m, n \in \mathbb{Z}\}$ the correspondence from G to $Z \oplus Z$ given by $\phi(3^m 9^n) = (m, n)$ is not well-defined since $\phi(3^2 9^0) \neq \phi(3^0 9^1)$ and $3^2 9^0 = 9 = 3^0 9^1$.
38. $|a_i| = \infty$ for some i .
39. Both D_6 and $D_3 \oplus Z_2$ have 1 element of order 1, 7 of order 2, 2 of order 3, and 2 of order 6.
40. 4
41. $U_5(35) = \{1, 6, 11, 16, 26, 31\}$; $U_7(35) = \{1, 8, 22, 29\}$.
42. Observe that $U(40) \oplus Z_6 \approx U(8) \oplus U(5) \oplus Z_6 \approx Z_2 \oplus Z_2 \oplus Z_4 \oplus Z_6$ and $U(72) \oplus Z_4 \approx U(9) \oplus U(8) \oplus Z_4 \approx Z_6 \oplus Z_2 \oplus Z_2 \oplus Z_4$ so they are isomorphic.
43. C^* has only one element of order 2 whereas $Z_2 \oplus Z_2$ has three elements of order 2.
44. If exactly one n_i is even then x is the unique element of order 2. Otherwise x is the identity.
45. Each cyclic subgroup of order 6 has two elements of order 6. So, the 24 elements of order 6 yield 12 cyclic subgroups of order 6. In general, if a group has $2n$ elements of order 6 it has n cyclic subgroups of order 6. (Recall from the Corollary of Theorem 4.4 if a group has a finite number of elements of order 6 the number is divisible by $\phi(6) = 2$).
46. $Z \oplus D_4 \oplus A_4$.
47. $\text{Aut}(U(25)) \approx \text{Aut}(Z_{20}) \approx U(20) \approx U(4) \oplus U(5) \approx Z_2 \oplus Z_4$.
48. S_3
49. In each position we must have an element of order 1 or 2 except for the case that every position has the identity. So, there are $2^k - 1$ choices. For the second question, we must use the identity in every position for which the order of the group is odd. So, there are $2^t - 1$ elements of order 2 where t is the number of $n_1, n_2, \dots, n_k$ that are even.
50. $Z_{10} \oplus Z_{12} \oplus Z_6 \approx Z_2 \oplus Z_5 \oplus Z_{12} \oplus Z_6 \approx Z_2 \oplus Z_{60} \oplus Z_6 \approx Z_{60} \oplus Z_6 \oplus Z_2$. $Z_{10} \oplus Z_{12} \oplus Z_6$ has 7 elements of order 2 whereas $Z_{15} \oplus Z_4 \oplus Z_{12}$ has only 3.

51. $\phi(18) = 6$; 0 ($Z_2 \oplus Z_3 \oplus Z_3$ is not cyclic).
52. Since $\phi(2, 3) = 2$ we have $8\phi(2, 3) = 16 = 1$. So, $1 = \phi(16, 24) = \phi(1, 4)$.
53. Since $(2, 0)$ has order 2, it must map to an element in Z_{12} of order 2. The only such element in Z_{12} is 6. The isomorphism defined by $(1, 1)x \rightarrow 5x$ with $x = 6$ takes $(2, 0)$ to 6. Since $(1, 0)$ has order 4, it must map to an element in Z_{12} of order 4. The only such elements in Z_{12} is 3 and 9. The first case occurs for the isomorphism defined by $(1, 1)x \rightarrow 7x$ with $x = 9$ (recall $(1, 1)$ is a generator of $Z_4 \oplus Z_3$); the second case occurs for the isomorphism defined by $(1, 1)x \rightarrow 5x$ with $x = 9$.
54. $U_4(140) \approx U(35) \approx U(5) \oplus U(7) \approx Z_4 \oplus Z_6$.
55. Since $a \in Z_m$ and $b \in Z_n$, we know that $|a|$ divides m and $|b|$ divides n . So, $|(a, b)| = \text{lcm}(|a|, |b|)$ divides $\text{lcm}(m, n)$.
56. Map $ax^2 + bx + c$ to (a, b, c) . In general, $\{a_{n-1}x^{n-1} + \cdots + a_0 \mid a_{n-1}, \dots, a_0 \in Z_m\}$ under addition modulo m is isomorphic to $Z_m \oplus \cdots \oplus Z_m$ (n copies).
57. Up to isomorphism, Z is the only infinite cyclic group and it has 1 and -1 as its only generators. The number of generators of Z_m is $|U(m)|$ so we must determine those m such that $|U(m)| = 2$. First consider the case where $m = p^n$, where p is a prime. Then the number of generators is $p^{n-1}(p-1)$. So, if $p > 3$ we will have more than 2 generators. When $p = 3$ we must have $n = 1$. Finally, $|U(2^n)| = 2^{n-1} = 2$ only when $n = 2$. This gives us Z_3 and Z_4 . When $m = p_1 p_2 \cdots p_k$, where the p 's are distinct primes we have $|U(m)| = |U(p_1)| |U(p_2)| \cdots |U(p_k)|$. As before no prime can be greater than 3. So, the only case is $m = 2 \cdot 3 = 6$.
58. Identify A with $(0, 0)$, T with $(1, 1)$, G with $(1, 0)$ and C with $(0, 1)$. Then a string of length n of the four bases is represented by a string of 0s and 1s of length $2n$ and the complementary string of $a_1 a_2 \dots a_{2n}$ is $a_1 a_2 \dots a_{2n} + 11 \dots 1$.
59. Each subgroup of order p consists of the identity and $p-1$ elements of order p . So, we count the number of elements of order p and divide by $p-1$. In $Z_p \oplus Z_p$ every nonidentity element has order p so there are $(p^2 - 1)/(p - 1) = p + 1$ subgroups of order p .
60. $Z \oplus D_3$.
61. In $Z \oplus Z_2$ $|(1, 1)| = \infty, |(-1, 0)| = \infty, |(1, 1)(-1, 0)| = |(0, 1)| = 2$.
62. $U(165) \approx U(11) \oplus U(3) \oplus U(5) \approx Z_{10} \oplus Z_2 \oplus Z_4$.
63. $U(165) \approx U(15) \oplus U(11) \approx U(5) \oplus U(33) \approx U(3) \oplus U(55) \approx U(3) \oplus U(5) \oplus U(11)$.

64. Since $U(2^n)$ is isomorphic to $Z_{2^{n-2}} \oplus Z_2$ and $Z_{2^{n-2}}$ and Z_2 each have exactly one element of order 2, $U(2^n)$ has exactly three elements of order 2.
65. We use the fact that $\text{Aut}(Z_{105}) \approx U(105) \approx U(3) \oplus U(5) \oplus U(7) \approx Z_2 \oplus Z_4 \oplus Z_6$. In order for (a, b, c) to have order 6 we could have $|c| = 6$ and a and b have orders 1 or 2. So we have 2 choices for each of a, b , and c . This gives 8 elements. The only other possibility for (a, b, c) to have order 6 is for $|c| = 3$ and a and b have orders 1 or 2, but not both have order 1. So we have 3 choices for a and b together and 2 choices for c . This gives 6 more elements for a total of 14 in all.
66. Use the fact that $U(27) \approx Z_{18}$.
67. $U(900) \approx Z_2 \oplus Z_6 \oplus Z_{20}$ so the element of largest order is the $\text{lcm}(2, 6, 20) = 60$.
68. $U(p^m) \oplus U(q^n) = Z_{p^m - p^{m-1}} \oplus Z_{q^n - q^{n-1}}$ and both of these groups have even order. Now use Theorem 8.2.
69. Observe that $U(55) \approx U(5) \oplus U(11) \approx Z_4 \oplus Z_{10}$ and $U(75) \approx U(3) \oplus U(25) \approx Z_2 \oplus Z_{20} \approx Z_2 \oplus Z_5 \oplus Z_4 \approx Z_{10} \oplus Z_4 \approx Z_4 \oplus Z_{10}$.
70. $U(144) \approx U(16) \oplus U(9) \approx Z_4 \oplus Z_2 \oplus Z_6$;
 $U(140) \approx U(4) \oplus U(5) \oplus U(7) \approx Z_2 \oplus Z_4 \oplus Z_6$.
71. $U_{125}(1000) = \{1, 251, 501, 751\}$.
72. Observe that $Z_2 \oplus Z_4 \oplus Z_9 \approx Z_4 \oplus Z_2 \oplus Z_9 \approx Z_4 \oplus Z_{18} \approx U(5) \oplus U(27) \approx U(135)$.
73. Since $U(pq) \approx U(p) \oplus U(q) \approx Z_{p-1} \oplus Z_{q-1}$ it follows that $k = \text{lcm}(p-1, q-1)$.
74. $U_{50}(200) = \{1, 51, 101, 151\}$ has order 4 whereas $U(4)$ has order 2. This is not a contradiction to Theorem 8.3 because 50 and $200/50 = 4$ are not relatively prime.
75. $|U(200)| = 80$; $|U(50) \oplus U(4)| = 40$.
76. Observe that $U(100) \approx U(4) \oplus U(25) \approx Z_2 \oplus Z_{20}$ so $n = \text{lcm}(2, 20) = 20$.
77. $U_8(40) \approx U(5) \approx Z_4$.
78. In the first case there are $2^k - 1$; in the second case there are $2^{k+2} - 1$.
79. None. Because $\text{gcd}(18, 12) = 6$, Step 3 of the Sender part of the algorithm fails.
80. Since $5 \cdot 29 = 1 \pmod{36}$, we have that $s = 29$. So, we need to compute $34^{29} \pmod{2701}$. The result is 1415, which converts to NO.

81. Because the block 2505 exceeds the modulus 2263, sending $2505^e \bmod 2263$ is the same as sending $242^e \bmod 2263$ which decodes as 242 instead of 2505.

CHAPTER 9

Normal Subgroups and Factor Groups

1. No, $(13)(12)(13)^{-1} = (23)$ is not in H .
2. For every α in S_n , $\alpha A_n \alpha^{-1}$ is even.
3. $HR_{90} = R_{270}H$; $DR_{270} = R_{90}D$; $R_{90}V = VR_{270}$
4. Solving $(12)(13)(14) = \alpha(12)$ for α we have $\alpha = (12)(13)(14)(12)$. Solving $(1234)(12)(23) = \alpha(1234)$ for α we have $\alpha = (234)$.
5. Say $i < j$ and $h \in H_i \cap H_j$. Then $h \in H_1 H_2 \cdots H_{j-1} \cap H_j = \{e\}$.
6. No. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. Then A is in H and B is in $GL(2, \mathbf{R})$ but BAB^{-1} is not in H .
7. H contains the identity so H is not empty. Let $A, B \in H$. Then $\det(AB^{-1}) = (\det A)(\det B)^{-1} \in K$. This proves that H is a subgroup. Also, for $A \in H$ and $B \in G$ we have $\det(BAB^{-1}) = (\det B)(\det A)(\det B)^{-1} = \det A \in K$ so $BAB^{-1} \in H$.
8. If k divides n , then $\langle k \rangle / \langle n \rangle$ is a cyclic group of order n/k . So it is isomorphic to $Z_{n/k}$.
9. Let $x \in G$. If $x \in H$, then $xH = H = Hx$. If $x \notin H$, then xH is the set of elements in G , not in H and Hx is also the elements in G , not in H .
10. **a.** $\alpha_5 H \neq H\alpha_5$. **b.** It proves that coset multiplication is not a binary operation.
11. Let $G = \langle a \rangle$. Then $G/H = \langle aH \rangle$.
12. Note that $aHbH = abH = baH = bHaH$.
13. in H .
14. 4
15. Since $(4U_5(105))^2 = 16U_5(105) = U_5(105)$, $|4U_5(105)| = 2$.
16. 3
17. $H = \{0 + \langle 20 \rangle, 4 + \langle 20 \rangle, 8 + \langle 20 \rangle, 12 + \langle 20 \rangle, 16 + \langle 20 \rangle\}$.
 $G/H = \{0 + \langle 20 \rangle + H, 1 + \langle 20 \rangle + H, 2 + \langle 20 \rangle + H, 3 + \langle 20 \rangle + H\}$

18. $60/4 = 15$
19. Observe that in a group G if $|a| = 2$ and $\{e, a\}$ is a normal subgroup then $xax^{-1} = a$ for all x in G . Thus $a \in Z(G)$. So, the only normal subgroup of order 2 in D_n is $\{R_0, R_{180}\}$ when n is even.
20. Let $H = U_5(20)$. Then $U(20)/H = \{H, 3H, 9H, 7H\}$.
21. By Theorem 9.5 the group has an element a of order 3 and an element b of order 11. Because $(ab)^{33} = (a^3)^{11}(b^{11})^3 = ee = e$ we know that $|ab|$ divides 33. $|ab| \neq 1$ for otherwise $|a| = |b^{-1}| = |b|$. $|ab| \neq 3$ for otherwise $e = (ab)^3 = a^3b^3 = b^3$, which is false. $|ab| \neq 11$ for otherwise $e = (ab)^{11} = a^{11}b^{11} = a^2$, which is false. So, $|ab| = 33$.
22. ∞ ; no, $(1, 1) + \langle(2, 2)\rangle$ has order 2 whereas in an infinite cyclic group every nonidentity element has infinite order.
23. $|G_1||G_2|/|H_1||H_2|$.
24. $Z_4 \oplus Z_2$. To see that there is no element of order 8 in the factor group observe that for any element (a, b) in $Z_4 \oplus Z_{12}$, $(a, b)^4 = (4a, 4b)$ belongs to $\{(0, 0), (0, 4), (0, 8)\} \in \langle(2, 2)\rangle$. So, the order of every element in the factor group divides 4. This rules out Z_8 . By observation, $(1, 0)\langle(2, 2)\rangle$ has order 4, which rules out $Z_2 \oplus Z_2 \oplus Z_2$.
25. Since the element $|3H|$ of G/H has order 4, $G/H \not\cong Z_2 \oplus Z_2 \oplus Z_2$. Because $9H$ and $31H$ have order 2, $G/H \not\cong Z_8$.
26. $Z_4 \oplus Z_2$; $\langle 17 \rangle \times \langle 41 \rangle$.
27. Since H and K have order 2 they are both isomorphic to Z_2 and therefore isomorphic to each other. Since $|G/H| = 4$ and $|3H| = 4$ we know that $G/H \approx Z_4$. On the other hand, direct calculations show that each of the three nonidentity elements in G/K has order 2, so $G/K \approx Z_2 \oplus Z_2$.
28. $Z_2 \oplus Z_2$; Z_4 .
29. Observe that nontrivial proper subgroups of a group of order 8 have order 2 or 4 and therefore are Abelian. Then use Theorem 9.6 and Exercise 4 of Chapter 8.
30. $U(165) = U_{15}(165) \times U_{11}(165) = U_{33}(165) \times U_5(165) = U_{55}(165) \times U_3(165)$.
31. Certainly, every nonzero real number is of the form $\pm r$, where r is a positive real number. Real numbers commute, and $\mathbf{R}^+ \cap \{1, -1\} = \{1\}$.
32. By Corollary 4 of Theorem 7.1, $x^m N = (xN)^m = N$, so $x^m \in N$.
33. In the general case that $G = HK$ there is no relationship. If $G = H \times K$, then $|g| = \text{lcm}(|h|, |k|)$ provided the $|h|$ and $|k|$ are finite. If $|h|$ or $|k|$ is infinite, so is $|g|$.

34. Since $1 = 3 \cdot 5 + (-2)7$, $m = (3m)5 + (-2m)7$. No, since $35 \in H \cap K$.
Suppose H is any subgroup of index 2. Then, since $|\mathbf{R}^*/H| = 2$, we have $a^2 \in H$ for all $a \in \mathbf{R}^*$. This implies that $\mathbf{R}^+ \subseteq H$. If there is some $a \in H$ with $a < 0$, then since $-a \in H$ we have $a^{-1}(-a) = -1 \in H$ also. But this implies that $\mathbf{R}^* \subseteq H$.
35. For the first question, note that $\langle 3 \rangle \cap \langle 6 \rangle = \{1\}$ and $\langle 3 \rangle \langle 6 \rangle \cap \langle 10 \rangle = \{1\}$.
For second question, observe that $12 = 3^{-1}6^2$ so $\langle 3 \rangle \langle 6 \rangle \cap \langle 12 \rangle \neq \{1\}$.
36. Certainly, $\mathbf{R}^+$ has index 2. Suppose that H has index 2 and is not $\mathbf{R}^+$.
Then $|\mathbf{R}^*/H| = 2$. So, for every nonzero real number a we have $(aH)^2 = a^2H = H$. Thus the square of every real number is in H . This implies that H contains all positive real numbers. Since H is not $\mathbf{R}^+$, it must contain some negative real number a . But then H contains $a\mathbf{R}^+$, which is the set of all negative real numbers. This shows that $H = \mathbf{R}^*$.
37. Say $|g| = n$. Then $(gH)^n = g^nH = eH = H$. Now use Corollary 2 to Theorem 4.1.
38. Observe that $(1/n + Z)$ has order n .
39. Let x belong to G and h belong to H . Then $xhx^{-1}H = (xh)x^{-1}H = (xh)Hx^{-1}H = xhHx^{-1}H = xHx^{-1}H = xH = H$, so xhx^{-1} belongs to H .
40. Observe that $(13)H(23)H = \{(132), (23), (12)\}$ whereas $(13)(23)H = \{(13), (123)\}$.
41. Suppose that H is a proper subgroup of Q of index n . Then Q/H is a finite group of order n . By Corollary 4 of Theorem 7.1 we know that for every x in Q we have nx is in H . Now observe that the function $f(x) = nx$ maps Q onto Q . So, $Q \subseteq H$.
42. Let $g \in G$. Then there is an element b in G so that $gH = (bH)^2 = b^2H$.
Thus $g = b^2h$ for some $h \in H$. But there is a c in H such that $h = c^2$. So, $g = (bc)^2$. The proof is valid for any integer.
43. Take $G = Z_6$, $H = \{0, 3\}$, $a = 1$, and $b = 4$.
44. That ϕ is 1-1 and onto is obvious. Moreover,

$$\begin{aligned} \phi((h_1h_2 \cdots h_n)(h'_1h'_2 \cdots h'_n)) &= \phi(h_1h'_1h_2h'_2 \cdots h_nh'_n) = \\ (h_1h'_1, h_2h'_2, \dots, h_nh'_n) &= (h_1, h_2, \dots, h_n)(h'_1, h'_2, \dots, h'_n) = \\ \phi((h_1h_2 \cdots h_n)\phi(h'_1h'_2 \cdots h'_n)). \end{aligned}$$
45. G and the trivial subgroup are normal. By Lagrange's Theorem all other subgroups have order p or 2. By Exercise 9 of this chapter every subgroup of order p in a group of order $2p$ is normal.
46. By Example 14 of Chapter 3, $Z(D_{13})$ is the identity. Then by Theorem 9.4, D_{13} is isomorphic to $\text{Inn}(D_{13})$.

47. By Lagrange, $|H \cap K|$ divides both 63 and 45. If $|H \cap K| = 9$, then $H \cap K$ is Abelian by Theorem 9.7. If $|H \cap K| = 3$, then $H \cap K$ is cyclic by the Corollary of Theorem 7.1. $|H \cap K| = 1$, then $H \cap K = \{e\}$.
48. Use Example 3 in Chapter 8 and the G/Z Theorem (Theorem 9.3).
49. By Lagrange's Theorem, $|Z(G)| = 1, p, p^2$, or p^3 . By assumption, $|Z(G)| \neq 1$ or p^3 (for then G would be Abelian). So, $|Z(G)| = p$ or p^2 . However, the " G/Z " Theorem (Theorem 9.3) rules out the latter case.
50. Use the G/Z Theorem.
51. Suppose that K is a normal subgroup of G and let $gH \in G/H$ and $kH \in K/H$. Then $gHkH(gH)^{-1} = gHkHg^{-1}H = gkg^{-1}H \in K/H$. Now suppose that K/H is a normal subgroup of G/H and let $g \in G$ and $k \in K$. Then $gkg^{-1}H = gHkHg^{-1}H = gHkH(gH)^{-1} \in K/H$ so $gkg^{-1} \in K$.
52. Say $|aH|$ has finite order n . Then $H = (aH)^n = a^nH$ so that a^n is in H . But this implies that $|a^n|$ and therefore $|a|$ is finite. Thus $aH = H$.
53. Say H has an index n . Then $(\mathbf{R}^*)^n = \{x^n \mid x \in \mathbf{R}^*\} \subseteq H$. If n is odd, then $(\mathbf{R}^*)^n = \mathbf{R}^*$; if n is even, then $(\mathbf{R}^*)^n = \mathbf{R}^+$. So, $H = \mathbf{R}^*$ or $H = \mathbf{R}^+$.
54. a. Since 1 and -1 commutes with every element of the group H is normal in G .
- b.
- | | | | | |
|-----|-----|-----|-----|-----|
| | 1 | i | j | k |
| 1 | 1 | i | j | k |
| i | i | 1 | k | j |
| j | j | k | 1 | i |
| k | k | j | i | 1 |
55. By Exercise 9, we know that K is normal in L and L is normal in D_4 . But $VK = \{V, R_{270}\}$ whereas $KV = \{V, R_{90}\}$. So, K is not normal in D_4 .
56. $x(H \cap N)x^{-1} = xHx^{-1} \cap xNx^{-1} = H \cap N$. The same argument works for the intersection of any family of normal subgroups.
57. In S_3 , let $H = \{(1), (12)\}$ and $K = \{(1), (13)\}$. Then the set $HK = \{(1), (13), (12), (12)(13)\} = \{(1), (13), (12), (132)\}$ does not contain $(13)(12) = (123)$.
58. By Example 5, NM is a subgroup. Also

$$xNM = (xN)M = (Nx)M = N(xM) = N(Mx) = NMx.$$

59. Let $H = \langle a^k \rangle$ be any subgroup of $N = \langle a \rangle$. Let $x \in G$ and let $(a^k)^m \in H$. We must show that $x(a^k)^m x^{-1} \in H$. Note that $x(a^k)^m x^{-1} = x(a^{km})x^{-1} = (xax^{-1})^{km} = (a^r)^{km} = (a^k)^{rm} \in \langle a^k \rangle$. (Here we used the normality of N to replace xax^{-1} by a^r .)

60. Use Theorem 9.4.
61. $\gcd(|x|, |G|/|H|) = 1$ implies $\gcd(|xH|, |G/H|) = 1$. But $|xH|$ divides $|G/H|$. Thus $|xH| = 1$ and therefore $xH = H$.
62. a. Observe that for any g in G ,

$$g(x^{-1}y^{-1}xy)g^{-1} = (gx^{-1}g^{-1})(gy^{-1}g^{-1})(gxyg^{-1})(gyg^{-1}) \in S.$$

- b. Observe that $xG'yG' = yG'xG'$ if and only if $x^{-1}y^{-1}xy \in G'$.
- c. Observe that $xNyN = yNxN$ implies $x^{-1}y^{-1}xyN = N$. Thus $x^{-1}y^{-1}xy \in N$.
- d. Let $h \in H$ and $g \in G$. Then $ghg^{-1}h^{-1} \in G' \leq H$ so that $ghg^{-1} \in Hh = H$.
63. Observe that for every positive integer n , $(1+i)^n$ is not a real number. So, $(1+i)\mathbf{R}^*$ has infinite order.
64. Let C the collection of all subgroups of G of order n . Then, since $|H| = n$ implies $|xHx^{-1}| = n$, we have

$$x(\cap_{H \in C} H)x^{-1} = \cap_{H \in C} xHx^{-1} = \cap_{H \in C} H.$$

65. Suppose that $\text{Aut}(G)$ is cyclic. Then $\text{Inn}(G)$ is also cyclic. So, by Theorem 9.4, G/Z is cyclic and from Theorem 9.3 it follows that G is Abelian. This is a contradiction.
66. Let $x \in K$. Then $|x| = p^i$ for some i . So $(xH)^{p^i} = x^{p^i}H = H$ and, because $|G/H| = m$, also we have $(xH)^m = H$. Thus $|xH|$ divides both m and p^i . From $\gcd(p, m) = 1$, we have $|xH| = 1$ and therefore $xH = H$. This proves that $K \subseteq H$.
67. Say $|gH| = n$. Then $|g| = nt$ (by Exercise 37) and $|g^t| = n$. For the second part observe that every nonidentity element of Z has infinite order while $1 + \langle 3 \rangle$ has order 3 in $Z/\langle 3 \rangle$.
68. Suppose that H is a subgroup of S_4 of order 12 distinct from A_4 . Then Example 5 in this chapter and Theorem 7.2 $HA_4 = S_4$ and $|HA_4| = 12 \cdot 12/|H \cap A_4|$. It follows that $|H \cap A_4| = 6$. But this contradicts Example 5 of Chapter 7.
69. First note that $|G/Z(G)| = |G|/|Z(G)| = 30/5 = 6$. By Theorem 7.3, the only groups of order 6 up to isomorphism are Z_6 and D_3 . But $G/Z(G)$ can't be cyclic for if so, then by Theorem 9.3, G would be Abelian. In this case we would have $Z(G) = G$. If $|Z(G)| = 3$, then $|G/Z(G)| = 10$ and by Theorem 7.3 G is isomorphic to Z_{10} or D_5 . Theorem 9.3 rules out Z_{10} . If $|G| = 2pq$ where p and q are distinct odd primes and $|Z(G)| = p$ or q , then $G/Z(G)$ is isomorphic to D_q or D_p , respectively.

70. Let $g \in G$ and let $H = \{e, h\}$. Then $\{g, gh\} = gH = Hg = \{hg, g\}$. So, $gh = hg$.
71. If A_5 had a normal subgroup of order 2 then, by Exercise 70, the subgroup has a nonidentity element that commutes with every element of A_5 . An element of A_5 of order 2 has the form $(ab)(cd)$. But $(ab)(cd)$ does not commute with (abc) , which also belong to A_5 .
72. Since H has index 2 in G is it a normal subgroup of G and $|G/H| = 2$. It follows that for every a in G we have $(aH)^2 = H$. If a is an element of G of order $2n + 1$, then $H = a^{2n+1}H = ((aH)^2)^n aH = aH$. Thus, a is in H .